

Dividing a proper fraction by a natural number

Sharing a part into equal pieces — the denominator grows, the numerator does not.

LESSON 22–23

2 × 40 MIN

MATEMATIKA 6, §5

S7 7.4

LESSON CLOCK

20'

25'

20'

12'

3

Sharing a part	20 min
The rule	25 min
Cancelling first	20 min
Problems	12 min
Homework	3 min

LEARNING OBJECTIVES

- Divide a fraction by a whole number and explain the result with a picture.
- Write the division as multiplication by the reciprocal.
- Cancel before multiplying.
- Solve simple sharing problems with fractions.

TEXTBOOK

National	pp. 58–64
Cambridge	Stage 7, pp. 71–75
Workbook	Exercise 7.4

01

Explanation

Sharing a part

Half a cake shared between three people: each gets a sixth. The number of pieces the whole is cut into has trebled, and the number taken has not.

$\frac{1}{2} \div 3 = \frac{1}{6}$

DIVISION	IN WORDS	ANSWER
$\frac{1}{2} \div 3$	half shared among three	$\frac{1}{6}$
$\frac{2}{3} \div 2$	two thirds shared between two	$\frac{1}{3}$
$\frac{3}{4} \div 5$	three quarters among five	$\frac{3}{20}$
$\frac{4}{5} \div 4$	four fifths among four	$\frac{1}{5}$

DIVIDING MAKES THE PIECES SMALLER

The answer is always less than the fraction you started with. That is the check to make before writing anything down.

The rule

$$\frac{a}{b} \div n = \frac{a}{b \cdot n} = \frac{a}{b} \cdot \frac{1}{n}$$

Multiply the denominator by the whole number — or, in the form that will cover every case from now on, multiply by the reciprocal of the divisor.

DIVISION	AS MULTIPLICATION	ANSWER
$\frac{3}{4} \div 5$	$\frac{3}{4} \cdot \frac{1}{5}$	$\frac{3}{20}$
$\frac{5}{7} \div 2$	$\frac{5}{7} \cdot \frac{1}{2}$	$\frac{5}{14}$
$\frac{2}{9} \div 3$	$\frac{2}{9} \cdot \frac{1}{3}$	$\frac{2}{27}$

THE NUMERATOR IS UNTOUCHED

$\frac{3}{4} \div 5$ is $\frac{3}{20}$, not $\frac{15}{4}$ and not the whole fraction turned upside down. Only the denominator grows.

Cancelling first

When the whole number divides the numerator, cancel — the answer comes out in lowest terms with no extra step.

DIVISION	CANCEL	ANSWER
$\frac{4}{5} \div 4$	$4 \div 4 = 1$	$\frac{1}{5}$
$\frac{6}{7} \div 3$	$6 \div 3 = 2$	$\frac{2}{7}$
$\frac{9}{10} \div 3$	$9 \div 3 = 3$	$\frac{3}{10}$
$\frac{2}{3} \div 2$	$2 \div 2 = 1$	$\frac{1}{3}$

CANCELLING FIRST KEEPS THE NUMBERS SMALL

$\frac{9}{10} \div 3$ done without cancelling gives $\frac{9}{30}$, which then has to be simplified anyway. One step is better than two.

Problems

PROBLEM	WORKING	ANSWER
$\frac{3}{4}$ of a litre of juice shared between 3 glasses	$\frac{3}{4} \div 3$	$\frac{1}{4}$ litre each
$\frac{5}{6}$ of a field ploughed in 5 days, equally	$\frac{5}{6} \div 5$	$\frac{1}{6}$ a day
$\frac{7}{8}$ of a metre of ribbon cut into 7 pieces	$\frac{7}{8} \div 7$	$\frac{1}{8}$ m each
a bottle $\frac{4}{5}$ full poured into 2 equal jugs	$\frac{4}{5} \div 2$	$\frac{2}{5}$ each

“SHARED BETWEEN” MEANS DIVIDE

Every one of these problems contains that phrase or one like it. Finding it in the sentence is the whole difficulty; the arithmetic afterwards is one line.

02 Worked examples

EXAMPLE 1 Compute $\frac{3}{4} \div 5$.

① Multiply by the reciprocal of 5.

② $\frac{3}{4} \cdot \frac{1}{5}$

③ $= \frac{3}{20}$

Smaller than $\frac{3}{4}$ ✓

ANSWER

$$\frac{3}{20}$$

EXAMPLE 2 Compute $\frac{6}{7} \div 3$.

① 3 divides the numerator 6.

② Cancel: $6 \div 3 = 2$.

③ $= \frac{2}{7}$

ANSWER

$$\frac{2}{7}$$

EXAMPLE 3 $\frac{3}{4}$ of a litre of juice is shared equally between 3 glasses. How much is in each?

① “Shared between” means divide.

② $\frac{3}{4} \div 3$

③ Cancel the 3s.

④ $= \frac{1}{4}$ litre.

ANSWER

$\frac{1}{4}$ litre

03 Interactive model

Fold a paper strip in half, then cut the half into three; the sixth appears in the hand before it appears on the board.

Dividing a fraction by a whole number

$$\frac{3}{4} \div 5$$

① $\frac{3}{4} \cdot \frac{1}{5}$

Multiply
by
the
reciprocal.

② Nothing cancels:

3 and

5 share no factor.

③ $\frac{3}{20}$

Smaller
than
 $\frac{3}{4}$
—
as it
must
be.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To divide	Bo'lish	Делить
Divisor	Bo'luvchi	Делитель
Reciprocal	Teskari son	Обратное число
To share equally	Teng taqsimlash	Разделить поровну
Part	Ulush	Доля
Whole	Butun	Целое
To cancel	Qisqartirish	Сокращать
Lowest terms	Qisqarmas ko'rinish	Несократимый вид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

1 2 ÷ 3 equals:

$$\frac{1}{6}$$

$$\frac{3}{2}$$

$$\frac{1}{5}$$

$$\frac{2}{3}$$

Dividing a fraction by a whole number makes it:

larger

smaller

the same

negative

3 4 ÷ 5 equals:

$$\frac{15}{4}$$

$$\frac{3}{20}$$

$$\frac{3}{9}$$

$$\frac{4}{15}$$

6 7 ÷ 3 equals:

$$\frac{2}{7}$$

$$\frac{6}{21}$$

$$\frac{18}{7}$$

$$\frac{2}{21}$$

Dividing by n is the same as multiplying by:

$$n$$

$$\frac{1}{n}$$

$$-n$$

$$n^2$$

4 5 ÷ 4 equals:

$$\frac{1}{5}$$

$$\frac{4}{20}$$

$$\frac{16}{5}$$

$$\frac{1}{20}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{1}{2} \div 3$

ANSWER $\frac{1}{6}$

2. $\frac{2}{3} \div 2$

ANSWER $\frac{1}{3}$

3. $\frac{3}{4} \div 5$

ANSWER $\frac{3}{20}$

4. $\frac{4}{5} \div 4$

ANSWER $\frac{1}{5}$

5. $\frac{6}{7} \div 3$

ANSWER $\frac{2}{7}$

6. $\frac{5}{7} \div 2$

ANSWER $\frac{5}{14}$

7. $\frac{9}{10} \div 3$

ANSWER $\frac{3}{10}$

Medium · the standard question

7 PROBLEMS

1. $\frac{2}{9} \div 3$

ANSWER $\frac{2}{27}$

2. $\frac{8}{9} \div 4$

ANSWER $\frac{2}{9}$

3. $\frac{3}{4}$ of a litre into 3 glasses

ANSWER $\frac{1}{4}$ litre

4. $\frac{5}{6}$ of a field in 5 days

ANSWER $\frac{1}{6}$ a day

5. $\frac{7}{8}$ m of ribbon into 7 pieces

ANSWER $\frac{1}{8}$ m

6. $\frac{4}{5}$ of a bottle into 2 jugs

ANSWER $\frac{2}{5}$

7. $\frac{10}{11} \div 5$

ANSWER $\frac{2}{11}$

Hard · stretch and reason**7 PROBLEMS**

1. $\frac{5}{8} \div 10$

ANSWER $\frac{1}{16}$

2. $\frac{12}{13} \div 6$

ANSWER $\frac{2}{13}$

3. $1\frac{1}{2} \div 3$

ANSWER $\frac{1}{2}$

4. $2\frac{2}{3} \div 4$

ANSWER $\frac{2}{3}$

5. A rope $\frac{9}{10}$ m long is cut into 6 equal pieces

ANSWER $\frac{3}{20}$ m

6. Why is the answer always smaller than the fraction?

ANSWER The same part is cut into more pieces

7. Divide $\frac{3}{7}$ by 3 and then by 2

ANSWER $\frac{1}{14}$

07 Homework

Homework — 5 tasks

Cancel before multiplying, and check that the answer is smaller than you started with.

1. Compute $\frac{2}{5} \div 3$ and $\frac{3}{8} \div 2$.
2. Compute $\frac{8}{9} \div 2$ and $\frac{10}{13} \div 5$, cancelling first.
3. Compute $1\frac{1}{3} \div 4$.
4. $\frac{5}{6}$ of a litre of milk is shared between 5 cups. How much is in each?
5. Explain in one sentence why dividing a fraction by 4 makes it smaller.